PosturePlus: A Multiplicative Temporal Stability Framework for Real-Time Posture Monitoring

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***Abstract*—The prolonged sitting on a laptop, or sitting at a desk, can lead to deterioration of posture; however, it is not usually an immediate phenomenon, rather, it is a slow process as fatigue builds up and deviations become normalized over time. The majority of the currently existing vision models end with either stationary thresholds or binary classification models, but they do not consider the temporal persistence of instability. Consequently, chronic slouch behaviour may be left unnoticed when short-term variance appears steady.**

The paper presents PosturePlus, which is a temporally based posture stability framework constructed around a multiplicative Posture Stability Index (PSI). This formulation directly separates instantaneous posture quality from behavioural stability over time and structurally binds them such that chronic instability normalization is avoided. Our unified model incorporates variance, drift slope, exponentially decayed episode density, recovery duration, along with guarded adaptive recalibration.

The multiplicative PSI in a structured 10-minute controlled protocol showed a 16–18 point deeper degradation in PSI values during a sustained RED phase in comparison to an additive formulation.

In threshold-based comparisons, BAD exposure was overestimated by the static deviation classification systems (57.8%), but was reduced by PSI-aligned classification (27.9%), while offering near-perfect episode-level agreement (1.00 vs 0.78).

During a 27-minute natural usage session, mean PSI was 88.43 with a minimum of up to 59 during uninterrupted fatigue cycles, validating sensitivity to persistence-induced instability.

These results show that multiplicative temporal coupling offers greater sensitivity to chronic instability compared to additive or threshold-based aggregation, re-framing posture monitoring as a persistence-aware stability modeling problem rather than a categorical detection task.

***Index Terms*—Posture monitoring, ergonomic analysis, temporal stability modeling, fatigue detection, adaptive recalibration, computer vision.**

I. Introduction

Extended work involving the use of laptops has become normalized, with prolonged sitting often accompanied by forward head posture and asymmetric shoulder alignment. Such patterns increase cervical loading and are strongly associated with musculoskeletal discomfort [1], [2]. There is also experimental evidence indicating that even short-term computer-based sitting can induce measurable musculoskeletal and cognitive effects [3].

In practice, posture degradation rarely occurs as a spontaneous transition from "good" to "bad." Instead, minor deviations accumulate gradually as fatigue develops, resulting in progressive baseline drift. Similar drift effects have been recorded in motor-control and postural-regulation studies, where sustained tasks generate gradual positional shifts over time [4], [5]. These findings imply that posture instability is inherently temporal.

Recent vision-based systems use pose estimation frameworks such as OpenPose and MediaPipe for real-time skeletal landmark extraction [6], [7]. Several approaches demonstrate accurate posture classification and ergonomic evaluation using keypoint-based models [8]–[10]. However, most systems formulate posture evaluation as threshold-based deviation detection or discrete classification. While effective for detecting large instantaneous errors, such methods inadequately capture sustained but stable slouch behavior.

In related domains such as behavioral drift detection and adaptive activity monitoring, temporal persistence and gradual distributional shifts are explicitly modeled as indicators of instability [11], [12]. By contrast, posture monitoring systems largely emphasize deviation magnitude rather than persistence.

To address this structural limitation, we introduce PosturePlus, a temporally grounded posture stability framework. At its core is a multiplicative Posture Stability Index (PSI) that distinguishes between instantaneous posture quality and temporal stability, and applies structural coupling between them. This formulation prevents persistent degradation from being artificially normalized by short-term stability effects.

Temporal dynamics are modeled through rolling variance, drift slope estimation, exponentially decayed episode density, and recovery duration analysis within an integrated representation. A guarded adaptive recalibration mechanism further restricts baseline adjustment to verified stable intervals, preventing gradual normalization of chronic slouch posture.

Experimental evaluation under structured and natural conditions demonstrates that multiplicative coupling improves degradation consistency during sustained instability, reduces over-triggered BAD exposure relative to static threshold baselines, and enhances episode-level coherence.

The main contributions of this work are:

1) A multiplicative PSI formulation coupling posture quality and temporal stability.
2) Integration of variance, drift slope, decayed episode density, and recovery modeling within a unified fatigue-aware framework.

3) A guarded adaptive recalibration mechanism preventing chronic slouch normalization.
4) Quantitative comparison against additive aggregation and static threshold baselines.

II. Related Work

A. Ergonomics and Postural Risk

A continuous forward head posture, as well as an asymmetric alignment of the shoulder for a very long time, has been associated with an increase in cervical load and musculoskeletal discomfort [1], [2]. Within a single sitting session, measurable short-term musculoskeletal and cognitive effects are seen in controlled studies when prolonged computer-based sitting is done [3]. Studies show that postural deviation is a physiologically relevant instability instead of just a geometric variation which accumulates over time. Motor control research has documented that unintentional positional drift occurs during sustained tasks, even without explicit intent [4], [5]. This drift phenomenon shows that posture degradation is gradual and temporally structured instead of being discretely state-based. However, most ergonomic assessment frameworks rely on static evaluation metrics or snapshot-based scoring schemes.

B. Vision-Based Posture Monitoring

Frameworks such as OpenPose and MediaPipe have enabled scalable, camera-based real-time posture monitoring systems [6], [7]. These works have demonstrated effective posture classification using keypoint-based deviation metrics and machine learning models [8]–[10]. Some recent approaches incorporate temporal learning architectures such as LSTMs for ergonomic risk assessment during particular task-specific activities.

These systems often achieve high detection accuracy, and hence posture assessment is commonly formulated as threshold-based deviation or discrete classification. The magnitude of deviation is often evaluated relative to angular limits or baseline calibration. These systems, although effective for identifying large instantaneous errors, such formulation imperfectly captures sustained yet stable slouch behavior where short-term variance remains low.

C. Temporal Modeling and Drift-Aware Systems

In correlated domains, there is explicit mention of temporal persistence and distributional drift as indicators of behavioral instability. Behavioral drift detection in smart environments [11] and adaptive human activity recognition systems [12] integrate these temporal dynamics to detect gradual state transitions. Deep temporal models that are used for fatigue detection also emphasize progressive degradation patterns instead of isolated deviations [13].

These approaches then recognize that instability often emerges from accumulation rather than sudden threshold crossing. However, temporal modeling principles such as variance analysis, drift slope estimation, and exponential decay weighting are rarely unified among posture monitoring systems.

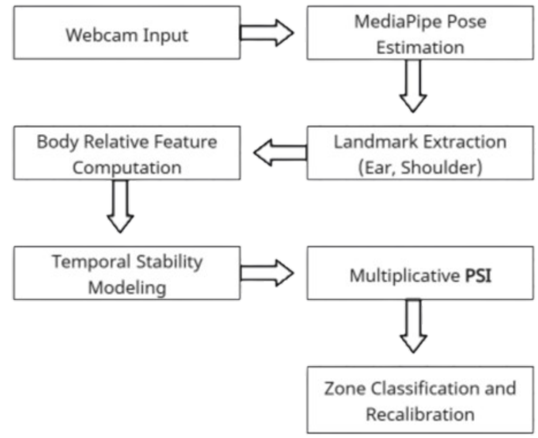

Fig. 1. High-level system architecture of PosturePlus. The pipeline consists of pose acquisition, body-relative feature extraction, temporal stability modeling, and multiplicative PSI computation with guarded recalibration.

D. Positioning of the Present Work

Posture monitoring systems that exist usually prioritize deviation magnitude over persistence. Additive aggregation or static thresholding may allow a stable but chronically degraded posture to remain insufficiently penalized. This structural gap is addressed by the presented work by explicitly separating instantaneous posture quality from temporal stability and coupling them multiplicatively within a unified Posture Stability Index (PSI).

The proposed framework re-frames the deviation-based detection to a persistence-aware stability modeling problem. By integrating variance, drift slope, decayed episode density, and guarded adaptive recalibration, PosturePlus aims to prevent the slow normalization of chronic slouch behavior along with preserving responsiveness to correction.

III. System Architecture

The PosturePlus framework follows a real-time processing pipeline consisting of four stages: (1) monocular pose acquisition, (2) body-relative feature extraction, (3) temporal stability modeling, and (4) PSI computation with adaptive recalibration. A high-level overview of the system architecture is illustrated in Fig. 1. The design emphasizes scale invariance, temporal persistence modeling, and prevention of baseline drift normalization.

A. Pose Acquisition

The system operates in real time using a monocular RGB webcam. Human pose landmarks are extracted using the MediaPipe Pose framework [7]. Upper-body landmarks including left and right shoulders and left and right ears are used for deviation modeling.

B. Body-Relative Feature Engineering

To ensure scale invariance, deviation metrics are normalized relative to shoulder width. Let (x_{LS}, y_{LS}), (x_{RS}, y_{RS}), (x_{LE}, y_{LE}), and (x_{RE}, y_{RE}) denote the 2D image coordinates of the left shoulder, right shoulder, left ear, and right ear respectively.

The shoulder midpoint is defined as:

$$y_{S_{\text{mid}}} = \frac{y_{LS} + y_{RS}}{2} \tag{1}$$

The ear midpoint is defined as:

$$y_{E_{\text{mid}}} = \frac{y_{LE} + y_{RE}}{2} \tag{2}$$

Shoulder width is defined as:

$$W = |x_{LS} - x_{RS}| \tag{3}$$

If $W = 0$, deviation metrics are set to zero to avoid numerical instability.

Forward deviation is computed as:

$$d_f(t) = \frac{y_{E_{\text{mid}}} - y_{S_{\text{mid}}}}{W} \tag{4}$$

Shoulder imbalance is defined as:

$$d_s(t) = \frac{|y_{LS} - y_{RS}|}{W} \tag{5}$$

Lateral tilt is defined as:

$$d_l(t) = |y_{LE} - y_{RE}| \tag{6}$$

Lateral tilt is not normalized by shoulder width, as it represents relative vertical asymmetry independent of global body scale. Each deviation signal is smoothed using an exponential moving average (EMA):

$$\tilde{d}(t) = \alpha\, d(t) + (1 - \alpha)\, \tilde{d}(t-1), \quad \alpha = 0.1 \tag{7}$$

C. Zone Classification and Hysteresis

A weighted posture score is defined as:

$$S(t) = w_f|d_f(t)| + w_l|d_l(t)| + w_s|d_s(t)| \tag{8}$$

Zone transitions are governed by temporal hysteresis constraints to prevent oscillatory classification under transient movements.

IV. Mathematical Formulation of the Posture Stability Index

Posture assessment requires modeling both instantaneous deviation magnitude and temporal persistence. The proposed Posture Stability Index (PSI) therefore separates posture evaluation into two components: *quality* and *stability*.

A. Instantaneous Posture Quality

Let $S(t)$ denote the weighted deviation score defined in Section III. Over a rolling window T_q, the mean deviation is

$$\bar{S}(t) = \frac{1}{T_q} \int_{t-T_q}^{t} S(\tau)\, d\tau. \tag{9}$$

The normalized quality component is

$$Q(t) = \text{clip}\left(1 - \frac{\bar{S}(t)}{C_q}, 0, 1\right), \tag{10}$$

where C_q represents the maximum tolerable deviation magnitude and $\text{clip}(\cdot)$ bounds the value within $[0, 1]$. Sustained deviation therefore directly reduces posture quality.

B. Temporal Stability Modeling

While $Q(t)$ captures magnitude, degradation also depends on temporal behavior. Four components are modeled:

Rolling Variance:

$$V(t) = \frac{\sigma_S(t)}{C_v}, \tag{11}$$

where $\sigma_S(t)$ is the standard deviation over window T_v.

Drift Slope:

$$\Delta(t) = \frac{\max(0, \beta_S(t))}{C_d}, \tag{12}$$

where $\beta_S(t)$ is the least-squares slope of $S(t)$.

Episode Density:

$$E(t) = \frac{1}{C_e} \sum_k e^{-\lambda(t - t_k)}. \tag{13}$$

Recovery Duration:

$$R(t) = \frac{\bar{r}(t)}{C_r}. \tag{14}$$

The aggregate stability penalty is

$$P_s(t) = w_v V(t) + w_d \Delta(t) + w_e E(t) + w_r R(t), \tag{15}$$

with stability component

$$S_{\text{stab}}(t) = \text{clip}(1 - P_s(t), 0, 1). \tag{16}$$

C. Multiplicative Stability Anchoring

An additive formulation such as

$$PSI_{\text{add}}(t) = 100 \cdot \frac{Q(t) + S_{\text{stab}}(t)}{2} \tag{17}$$

may attenuate persistent degradation when short-term stability remains high. To prevent normalization of chronic slouch behavior, the proposed PSI enforces multiplicative coupling:

$$PSI(t) = 100 \cdot Q(t) \cdot S_{\text{stab}}(t). \tag{18}$$

Under this formulation, if $Q(t) \to 0$, then $PSI(t) \to 0$ regardless of transient stability, which ensures that sustained deviation cannot be structurally masked.

V. Adaptive Recalibration Framework

Static baseline calibration may become suboptimal during long sessions due to gradual posture shifts or camera drift. However, naive recalibration risks in redefining degraded posture as the new baseline. To prevent such normalization, PosturePlus implements a guarded adaptive recalibration mechanism.

A. Baseline Update Rule

Let $b(t)$ denote the baseline deviation reference. When recalibration criteria are satisfied, the baseline is updated using a low-pass adaptation rule:

$$b_{\text{new}} = \beta\, x(t) + (1-\beta)\, b_{\text{old}}, \tag{19}$$

where $x(t)$ is the current deviation metric and $\beta \in (0,1)$. In practice, $\beta = 0.01$ ensures slow, conservative drift. This exponential smoothing prevents abrupt baseline shifts.

B. Guarded Recalibration Conditions

Baseline updates are permitted only if all of the following conditions hold:

1) Current posture zone is GREEN.
2) $PSI(t) > \theta_{high}$.
3) No RED episodes occurred within blocking interval T_{block}.
4) Session-level PSI slope is non-negative.
5) Stability is sustained for at least T_{stable} seconds.

These constraints prevent recalibration during sustained RED exposure, negative PSI trends, or elevated instability density. Consequently, degraded posture cannot be gradually redefined as neutral.

Because updates occur only under high $Q(t)$ and stable $S_{\text{stab}}(t)$, PSI continuity is preserved. Moreover, the small adaptation coefficient ensures

$$|b_{\text{new}} - b_{\text{old}}| \ll 1, \tag{20}$$

hence enforcing a conservative drift control while maintaining interpretability of the PSI formulation.

VI. Experimental Protocol

The proposed PSI framework was evaluated under both structured and natural usage conditions.

A. Controlled Protocol

A 10-minute controlled session was designed to simulate five posture phases: (1) upright stability, (2) gradual forward lean, (3) sustained slouch, (4) active recovery, and (5) post-recovery stabilization. Each phase was maintained for a fixed duration to induce measurable transitions in deviation magnitude and temporal stability.

B. Natural Usage Session

A 27-minute natural usage session was recorded without exposing the user to PSI trends. This setting captured realistic posture drift, fatigue progression, and spontaneous correction behavior.

C. Data Logging and Metrics

For each frame, forward, lateral, and shoulder deviations, zone classification (GREEN / YELLOW / RED), PSI value, and recalibration events were recorded. Session-level aggregation was performed over fixed intervals to compute mean PSI, PSI slope, Stability Degradation Index (SDI), and RED exposure ratio.

D. Fatigue Heuristic

Fatigue risk was flagged when any of the following conditions were satisfied:

$$\text{slope} < -\theta_s \;\text{ or }\; SDI > \theta_d \;\text{ or }\; R_{\text{red}} > \theta_r \tag{21}$$

where θ_s, θ_d, and θ_r denote predefined degradation thresholds.

E. Episode-Level Evaluation

Labeled GOOD and BAD segments were manually annotated and grouped into contiguous episodes. Majority zone prediction within each episode was compared to ground truth to compute episode-level accuracy, avoiding frame-level bias and emphasizing temporal consistency.

VII. Results

A. Controlled Protocol Results

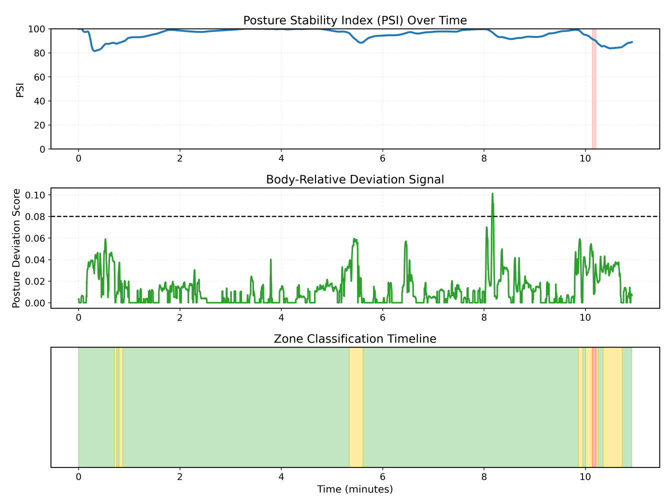

Fig. 2. Multi-panel visualization of system's behavior during the structured 10-minute controlled protocol. *Top:* PSI over the time showing progressive degradation during induced slouch phases and recovery upon correction. *Middle:* Body-relative deviation signal with dashed quality threshold. *Bottom:* Zone classification timeline; RED intervals align with sustained deviation and corresponding PSI decline.

Fig. 2 illustrates the PSI trajectory during the structured 10-minute protocol. A progressive decline in PSI is observed during the gradual lean and sustained slouch phases, reaching a minimum of approximately 60. The mean PSI during the slouch phase was reduced by more than 30% relative to the upright baseline. Upon active posture correction, PSI exhibits gradual recovery consistent with reduced deviation and episode density. No kind of artificial upward discontinuity was observed following recalibration, confirming the stability preservation.

B. Comparative Aggregation Analysis

The multiplicative PSI is compared against an additive aggregation model:

$$PSI_{\text{add}}(t) = 100 \cdot \frac{Q(t) + S_{\text{stab}}(t)}{2} \tag{22}$$

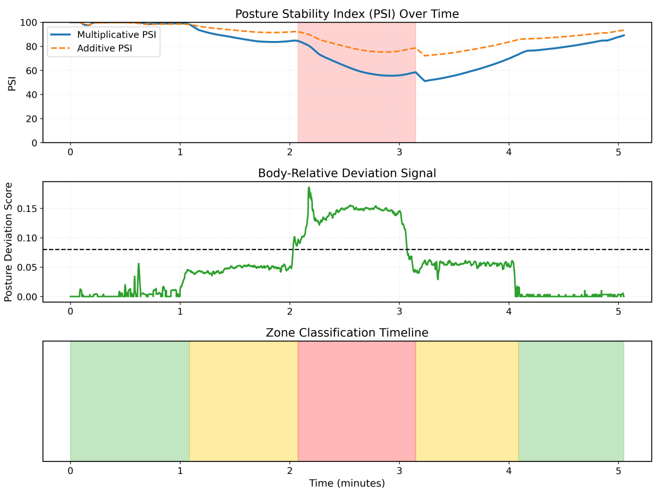

Fig. 3. Temporal comparison of multiplicative vs. additive PSI formulations during a representative controlled session. The additive formulation attenuates sustained instability relative to multiplicative coupling.

Fig. 3 shows that the additive model maintains elevated PSI values during sustained RED exposure, whereas the multiplicative formulation exhibits deeper degradation consistent with persistent instability. Three controlled protocol repetitions were performed under identical posture phase conditions; results are summarized in Table I.

TABLE I
MULTIPLICATIVE VS. ADDITIVE AGGREGATION COMPARISON (CONTROLLED PROTOCOL, 3 SESSIONS)

Metric	**Mult. (Mean ± Std)**	**Add. (Mean ± Std)**
Mean PSI (RED)	65.65 ± 2.28	81.69 ± 1.62
Min PSI (RED)	60.74 ± 4.56	78.66 ± 3.07
Recovery (PSI/min)	27.52 ± 6.92	15.77 ± 4.11

Across three repetitions, the multiplicative PSI remained approximately 16 points lower than the additive baseline during sustained RED exposure, demonstrating stronger sensitivity to chronic instability. The minimum PSI during RED phases showed an average gap of nearly 18 points, indicating deeper penalization of persistent slouch behavior. Recovery slope analysis further revealed steeper rebound dynamics in the multiplicative formulation.

C. Threshold Classification Baseline Comparison

To compare against conventional static classification approaches, a threshold-based baseline was evaluated using the posture deviation score without temporal modeling. Results are reported in Table II.

TABLE II
THRESHOLD VS. PSI CLASSIFICATION PERFORMANCE (CONTROLLED PROTOCOL, 3 SESSIONS)

Metric	**Threshold**	**PSI**
Episode Accuracy (Mean)	0.78	1.00
Mean BAD Ratio	57.8%	27.9%

Across three controlled repetitions, the threshold-based classifier achieved a mean episode accuracy of 0.78, compared to perfect episode consistency under the PSI framework. The threshold model predicted BAD posture in approximately 58% of frames, significantly overestimating sustained instability relative to the induced slouch duration. In contrast, the PSI predicted approximately 28% BAD exposure, closely aligned with the actual controlled slouch window.

D. Natural Session Results

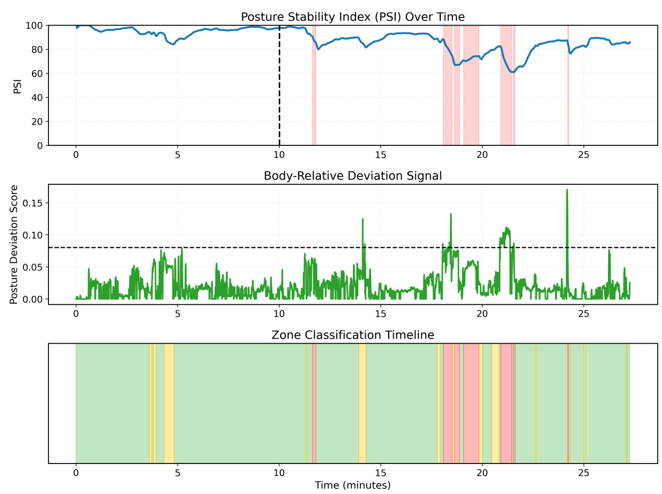

Fig. 4. Multi-panel visualization of the 27-minute natural usage session. *Top:* PSI evolution showing gradual mid-session degradation reaching a minimum near 60 during clustered RED exposure. *Middle:* Body-relative deviation signal with increased magnitude during fatigue phases. *Bottom:* Zone classification timeline illustrating temporally clustered RED episodes rather than isolated transient spikes.

Fig. 4 represents the PSI evolution over the 27-minute natural session. The mean PSI was approximately 88, with a minimum near 59 during the clustered RED exposure. Mid-session fatigue is characterized by increased RED exposure ratio and negative PSI slope. The computed SDI was positive and the fatigue heuristic was hence triggered. Recalibration only occurred once during a stable high-quality period, demonstrating correct enforcement of guard conditions. Table III summarizes the quantitative stability metrics across both sessions.

TABLE III
SESSION-LEVEL STABILITY METRICS (CONTROLLED VS. NATURAL USAGE)

Metric	**Controlled**	**Natural**
Session Duration (s)	641.32	1623.88
Mean PSI	95.38	88.43
Minimum PSI	80.83	59.56
PSI Slope (per second)	-0.0035	-0.0122
Stability Degradation Index	5.26	14.29
RED Exposure Ratio	0.007	0.085
Fatigue Flag	False	True

VIII. DISCUSSION

The results support the core claim that posture monitoring largely benefits from separating instantaneous deviation magnitude from temporal stability. The proposed framework models posture as a persistence-driven stability process instead of just a discrete state.

Multiplicative coupling enforces the structural interaction between posture quality and posture stability. While having sustained slouch phases, PSI exhibited consistent degradation without attenuation, unlike additive aggregation, which partially masked the chronic instability. This hence confirms that multiplicative anchoring preserves sensitivity to persistent deviation.

Fatigue behavior here emerged through temporally clustered RED exposure, negative PSI slope, and elevated SDI values, which indicated degradation development through accumulation instead of isolated spikes. With the integration of variance, drift slope, episode density decay, and recovery modeling in a unified system, the framework captures both magnitude and persistence dynamics.

Over-triggering behavior was demonstrated by threshold-based classification, whereas PSI-aligned classification reduces excessive BAD exposure while maintaining temporal consistency. These findings show that posture monitoring systems should prioritize persistence-aware stability modeling over magnitude-only thresholding.

IX. LIMITATIONS

A. Monocular Geometric Constraints

The system relies on a 2D landmark estimation from a monocular RGB webcam to preserve scalability and hardware independence. Hence, a true 3D spinal curvature and sagittal-plane translation cannot be directly measured without multiple camera angles or sensors. Forward deviation is inferred through relative landmark displacement instead of absolute depth estimation .

B. Heuristic Parameter Selection

Normalization constants and degradation thresholds were selected empirically to preserve stability behavior. Even though consistent across experimental and natural sessions, systematic hyperparameter optimization or data-driven calibration may improve robustness.

C. Absence of Clinical Validation

The framework focuses on the aspect of behavioral stability in a posture monitoring tool rather than a medical diagnostic system. Formal validation against clinical ergonomic assessments remains future work.

D. Environmental Variability

Lighting conditions, camera placement, and hardware differences may influence pose estimation stability, as different users tend to adopt different setups. Although cameras are continuously improving over time, placement and hardware differences might remain significant sources of variability.

X. FUTURE WORK

The presented work establishes multiplicative temporal coupling as a robust mechanism for stability-driven posture monitoring, though several extensions remain for further investigations.

Future research will focus on largely increasing multi-user validation across diverse anthropometric profiles in order to evaluate generalization to a greater extent. Making it an open-source deployed model, we will focus on getting as many reviews as possible to navigate multiple directions.

Data-driven optimization of degradation thresholds and normalization constants may further improve adaptability across users. Additionally, incorporation of depth-aware pose estimation or multi-view geometry may enhance geometric fidelity beyond 2D landmark inference.

Lastly, longitudinal studies examining the correlation between PSI trends and clinical musculoskeletal assessments would strengthen its validation in ergonomic health contexts.

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